SURVEYING & LEVELLING · NOTE SL06

Errors in Chain Surveying — Sources, Mistakes and Corrections

Course topics: SWCE1.4 L7

Sources of error

Type	Examples
Human (personal)	Miscounting arrows, misreading, poor ranging
Instrumental	Wear and distortion of links, stretching of tapes
Natural	Temperature changes; wind (minor effect)

ICAR eCourse: the total error in a line has two parts — cumulative errors, which grow in proportion to the length L, and compensating errors, which grow only in proportion to √L.

Common mistakes (ICAR eCourse)

- Displacing an arrow and replacing it inaccurately — mark each chain length with a cross as well as the arrow.
- Adding or omitting a full chain length through miscounting or lost arrows — the leader starts with all ten arrows and the follower counts them.
- Reading a tally from the wrong end, e.g. booking 9.6 m for 10.4 m.

Wrong chain length

- A chain seldom keeps its exact length. Stretching makes it **too long**; kinks and bent links make it **too short**.
- **Check** it against a standard steel tape on a level floor.

Correcting a measured length

$$\text{Correct distance} = (L' / L) \times \text{observed distance}$$

- L′ = incorrect (actual) length of the chain; L = its nominal (standard) length.
- If the mean error of the chain is e: chain too long → L′ = L + e; chain too short → L′ = L − e.

> ICAR eCourse illustration: a 1280 m line measured with a 20 m chain that is 0.02 m too long. Correct length = (20.02/20) × 1280 = 1281.28 m.

Correcting a measured area

$$\text{Actual area} = (L' / L)^2 \times \text{observed area} = ((L \pm e)^2 / L^2) \times \text{observed area}$$

For a small e/L the square reduces to:

Chain	Approximate correction
Too long	Actual area = observed area × (1 + 2e/L)
Too short	Actual area = observed area × (1 − 2e/L)

An area plotted from a 20 m chain that was 0.10 m too long reads 2.50 ha. Exact: $(20.10/20)^2 \times 2.50 = 2.525$ ha. Approximate: $2.50 \times (1 + 2 \times 0.10/20) = 2.525$ ha.

Corrected formula. The source notes print the area factor as $(L' \pm e)^2/L^2$. Since L' already includes the error ($L' = L \pm e$), the factor is $(L'/L)^2 = (L \pm e)^2/L^2$ — the error must not be added twice.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. A line measured with a chain that is too long will be

(a) shorter than the true length
(b) longer than the true length
(c) correct
(d) unaffected

Q2. Correct length = ?

(a) (L/L′) × observed
(b) (L′/L) × observed
(c) observed − e
(d) observed × L

Q3. Area correction factor for a chain too long by e (small e/L) is about

(a) 1 + e/L
(b) 1 + 2e/L
(c) 1 − 2e/L
(d) $1 + e^2/L^2$

Q4. Compensating errors in chaining vary in proportion to

(a) L
(b) √L
(c) L^2
(d) 1/L

Q5. A 1280 m line measured with a 20 m chain 0.02 m too long is actually

(a) 1278.72 m
(b) 1280.02 m
(c) 1281.28 m
(d) 1282.56 m

Answers

Q1 — (a) shorter than the true length — *Each chain length covers more ground than booked, so the booked distance is too small.*
Q2 — (b) (L′/L) × observed
Q3 — (b) 1 + 2e/L
Q4 — (b) √L — *ICAR eCourse.*
Q5 — (c) 1281.28 m — *(20.02/20) × 1280.*